

feature_matrix_julia_allGO_new.tsv quality analysis

Genes: 150

GO terms: 1,665

Density: 0.0844

Median active terms/gene: 119.0

Median genes/term: 9.0

Singleton/rare terms support ≤ 2 : 335

Terms support ≤ 5 : 598

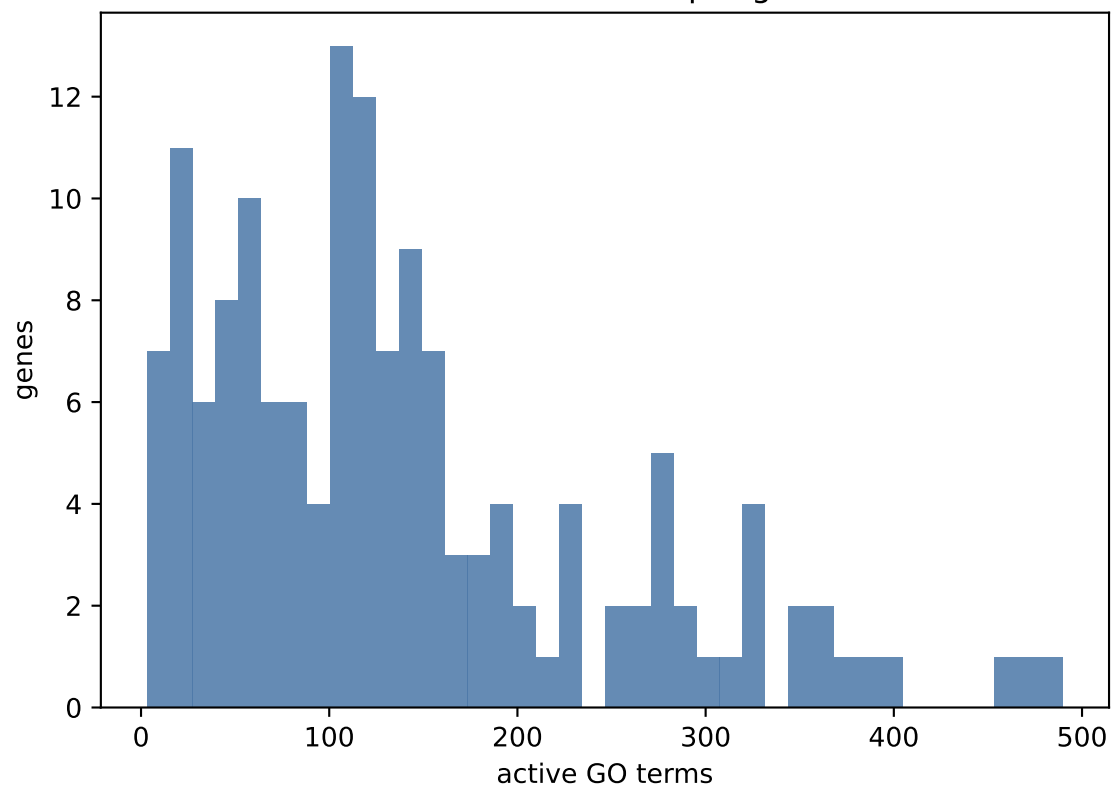
Terms present in $\geq 50\%$ genes: 9

Duplicate gene patterns: 0 groups

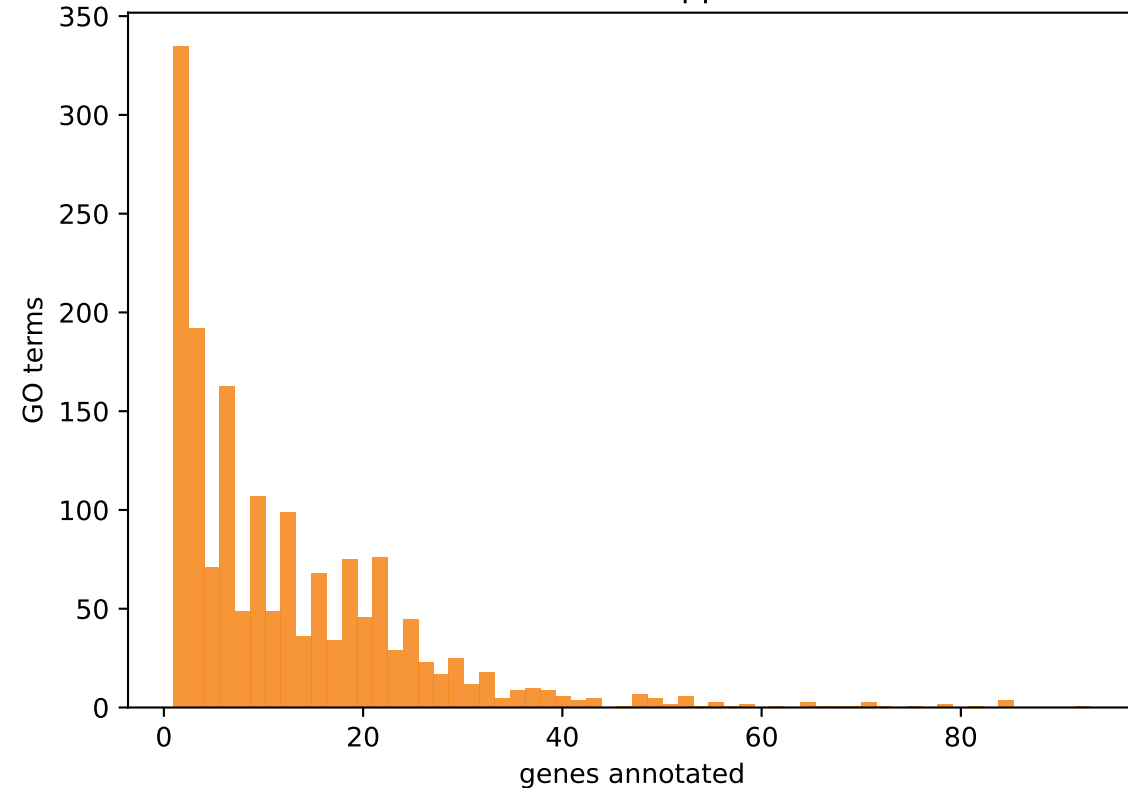
Duplicate GO-term patterns: 203 groups

Interpretation: sparse rare terms increase GO-IC overfitting risk, while broad terms can dominate distance geometry. Use this report to decide filtering/sensitivity grids.

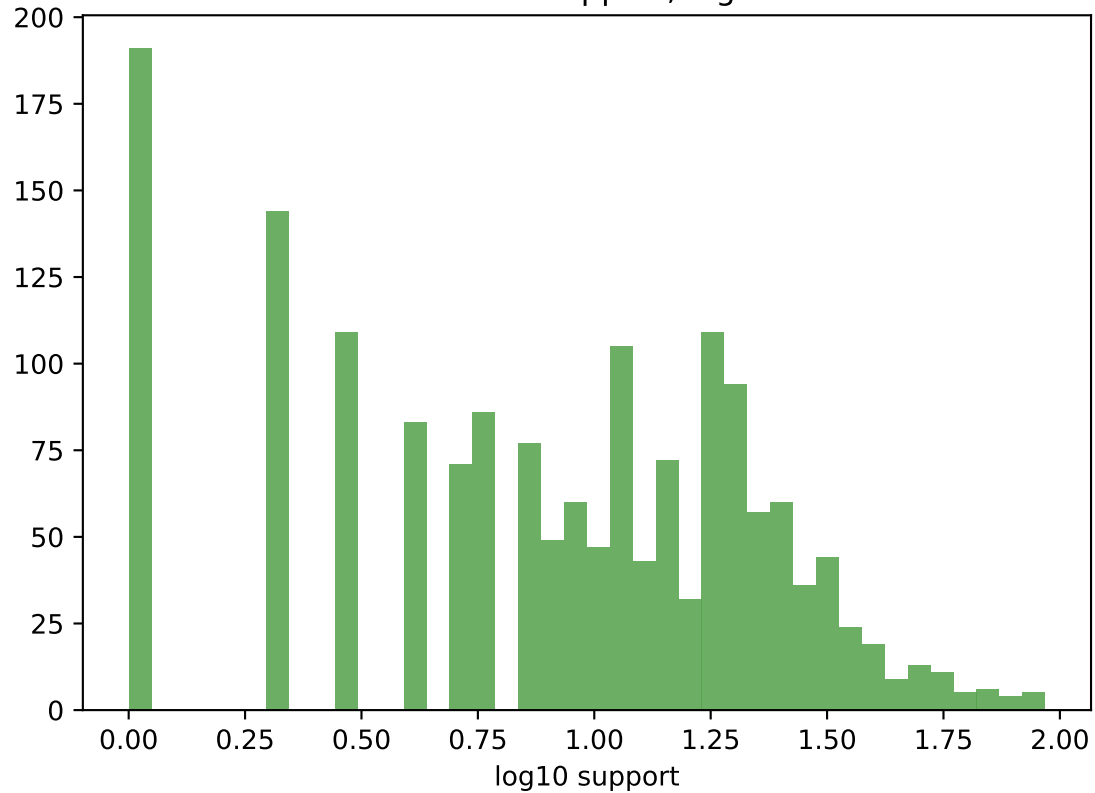
Annotation burden per gene



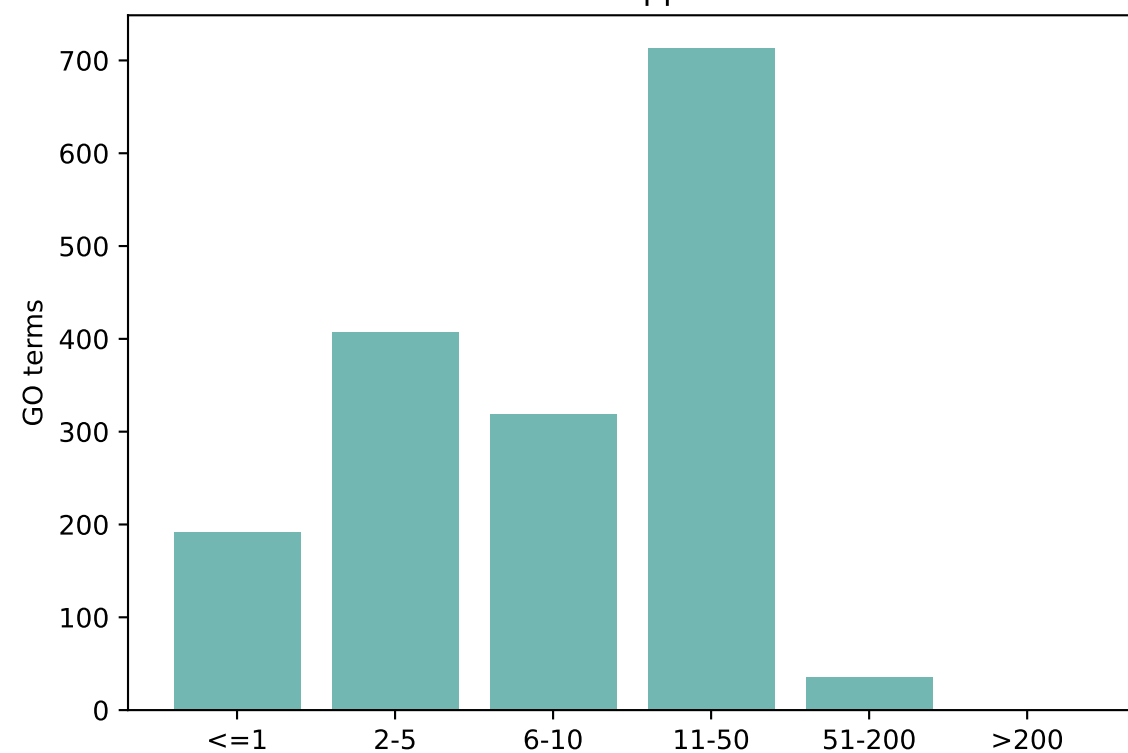
GO term support



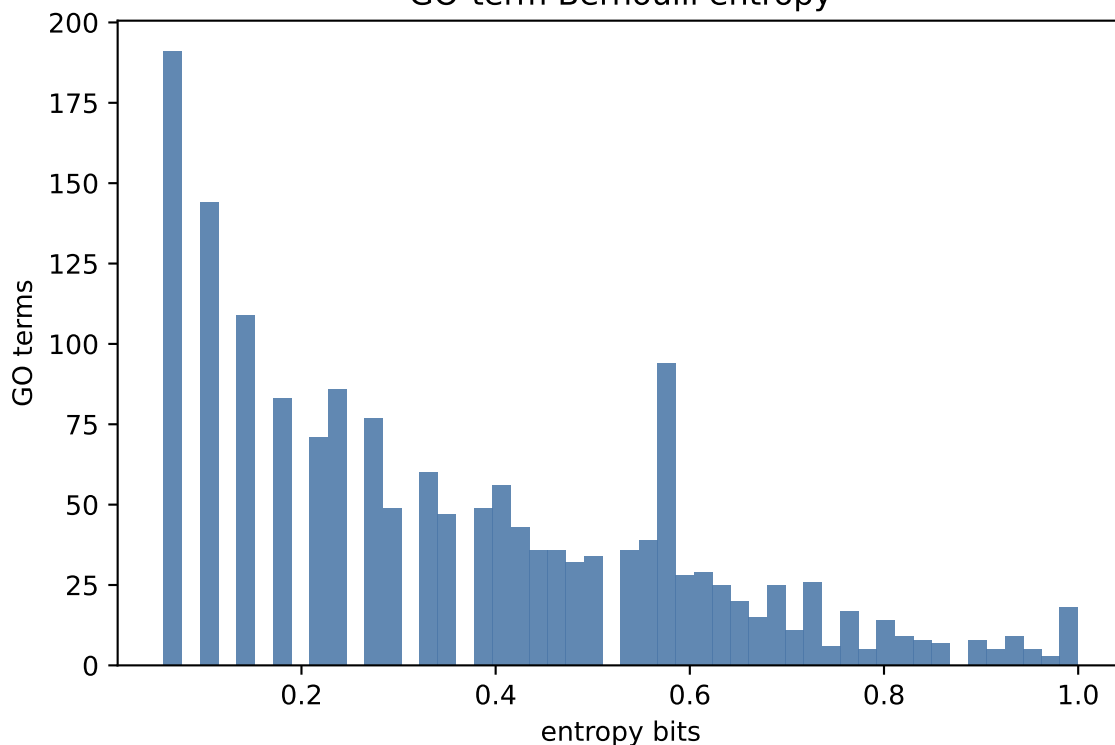
GO term support, log10



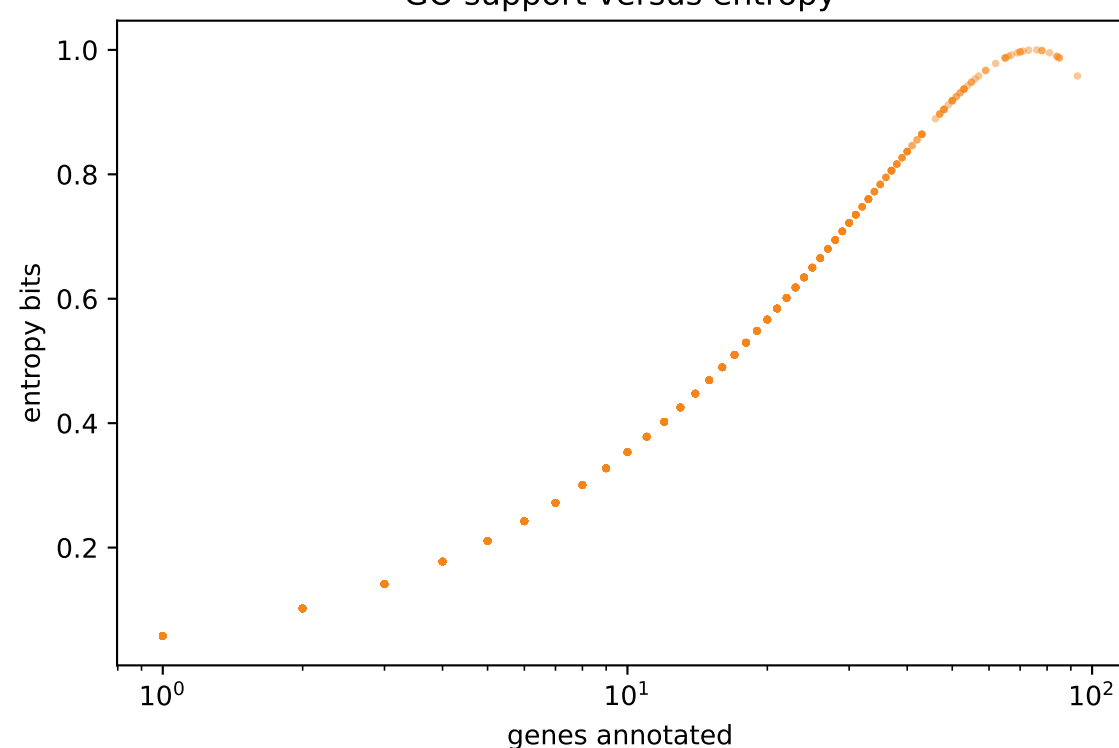
GO term support bins



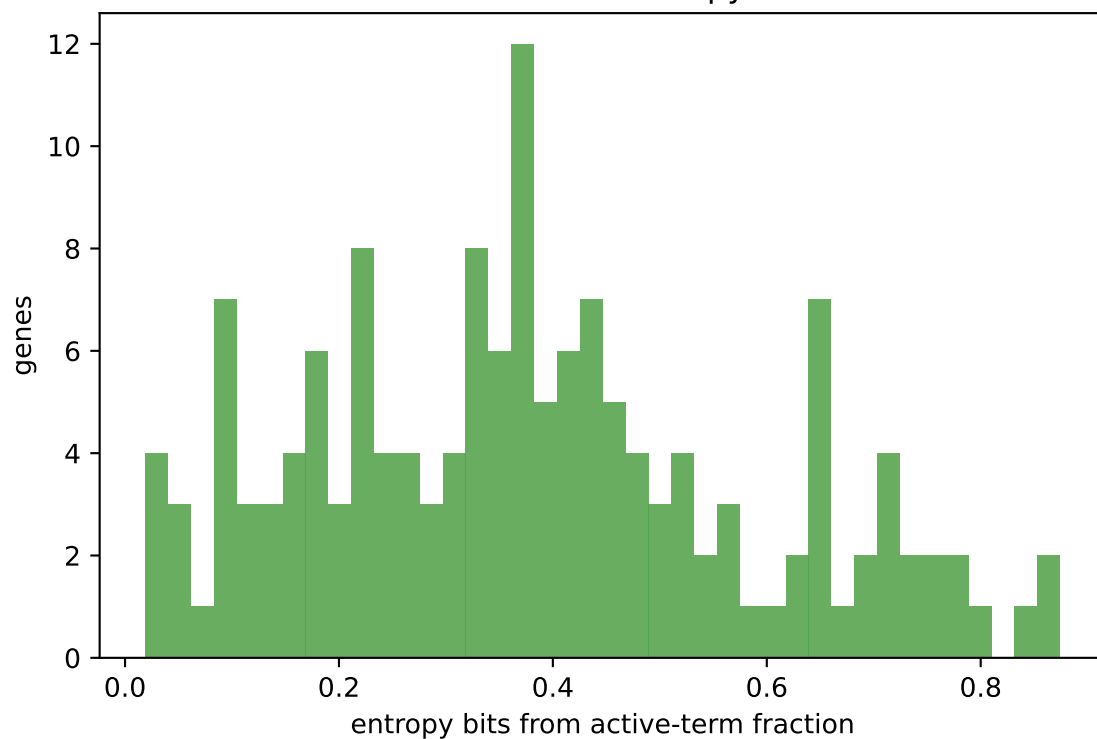
GO-term Bernoulli entropy



GO support versus entropy

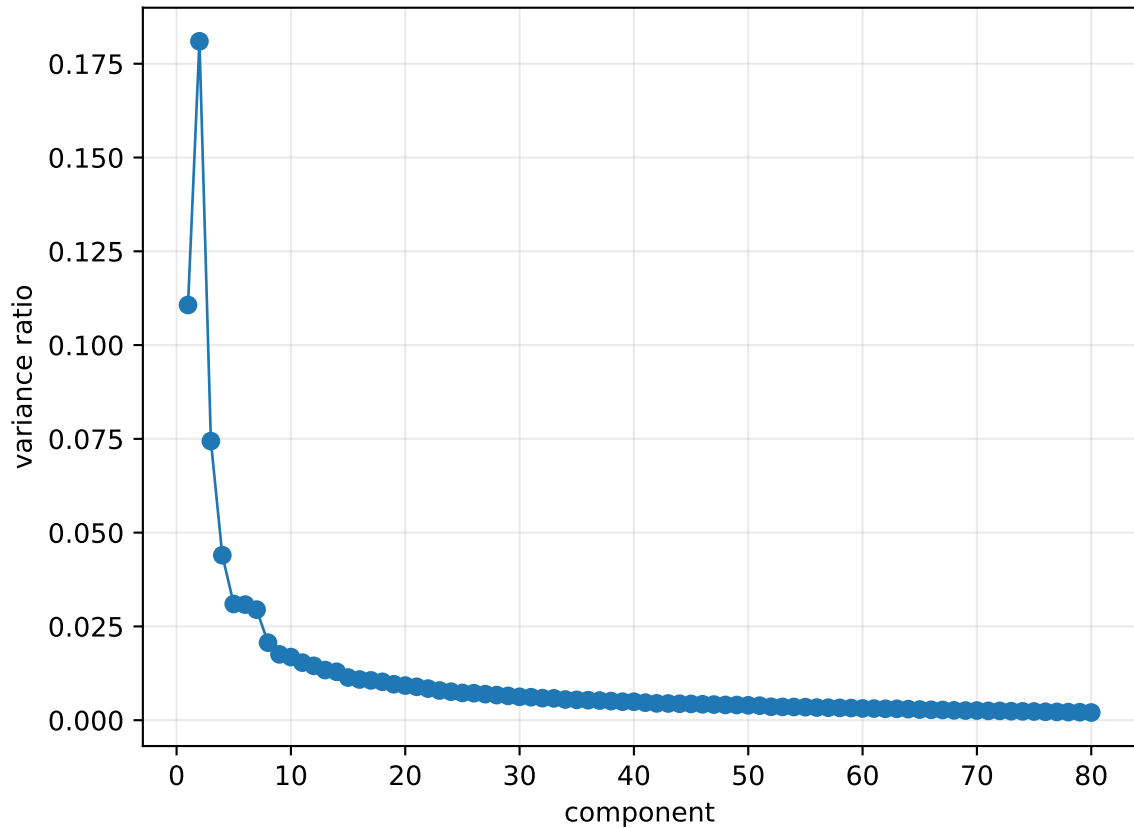


Gene row entropy

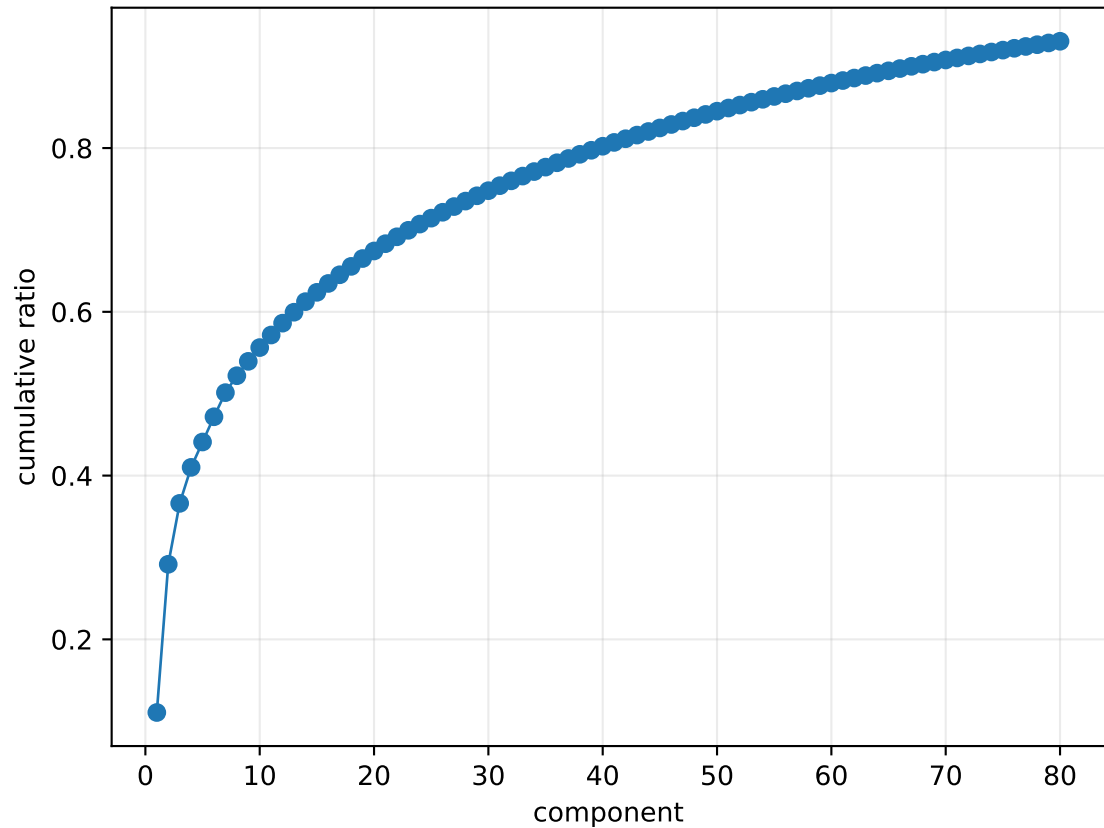


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Entropy summary
go_term_entropy_bits_mean: 0.360103
go_term_entropy_bits_median: 0.327445
go_term_entropy_bits_p25: 0.141441
go_term_entropy_bits_p75: 0.548225
go_term_entropy_bits_p95: 0.79504
go_term_entropy_bits_max: 0.999872
go_terms_entropy_le_0_05: 0
go_terms_entropy_le_0_10: 191
go_terms_entropy_le_0_20: 527
go_terms_entropy_ge_0_50: 496
go_terms_entropy_ge_0_80: 82
total_go_marginal_entropy_bits: 599.571
mean_go_marginal_entropy_bits_per_term: 0.360103
matrix_density_entropy_bits_per_cell: 0.417635
gene_row_entropy_bits_mean: 0.380364
gene_row_entropy_bits_median: 0.371391
gene_row_entropy_bits_p95: 0.745223
gene_row_entropy_bits_max: 0.874204
annotation_entry_entropy_across_genes_bits: 6.82711
annotation_entry_effective_genes: 113.544
annotation_entry_entropy_across_go_terms_bits: 10.0933
annotation_entry_effective_go_terms: 1092.38
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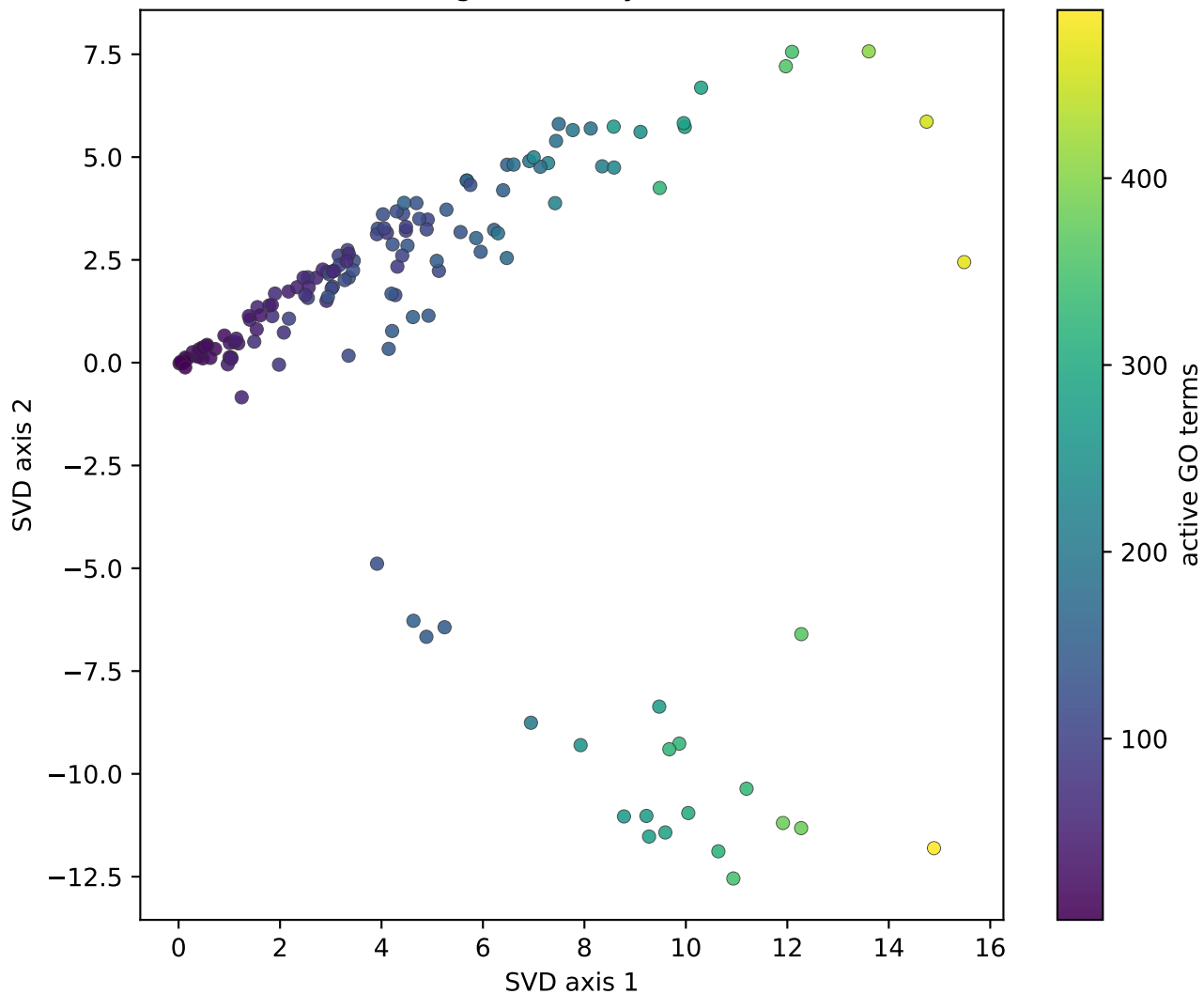
SVD explained variance ratio



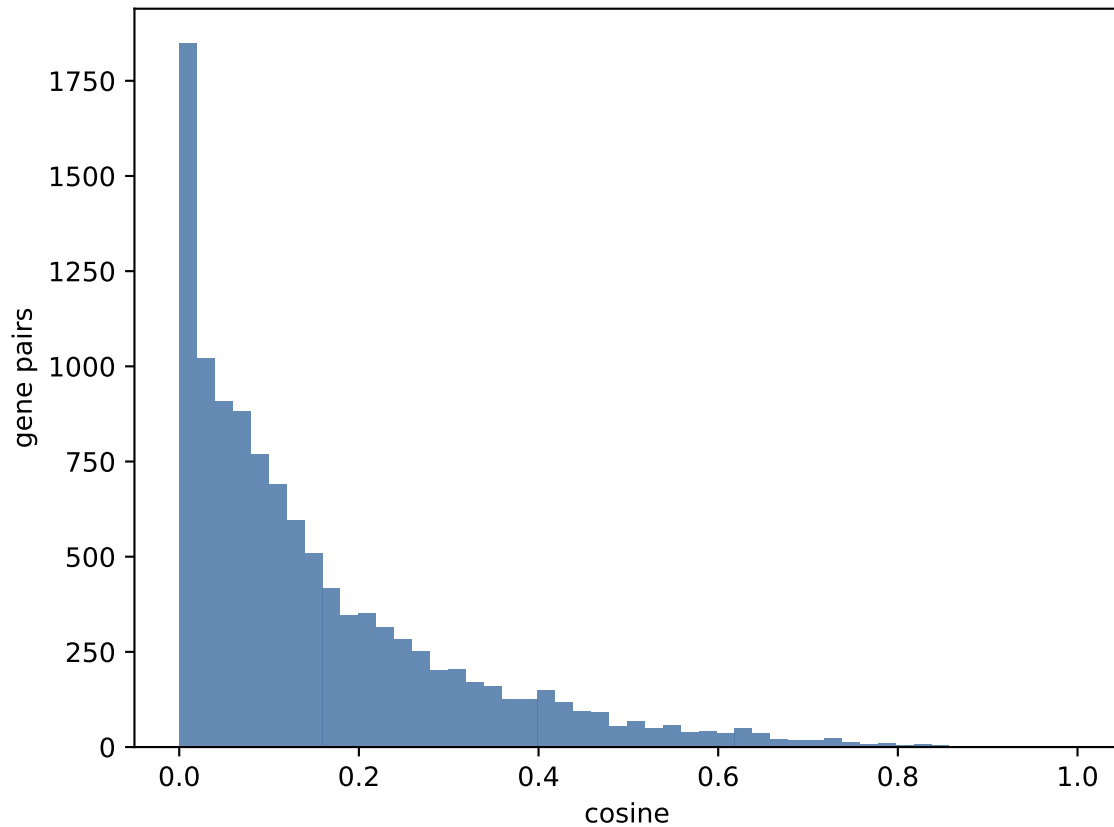
Cumulative SVD variance



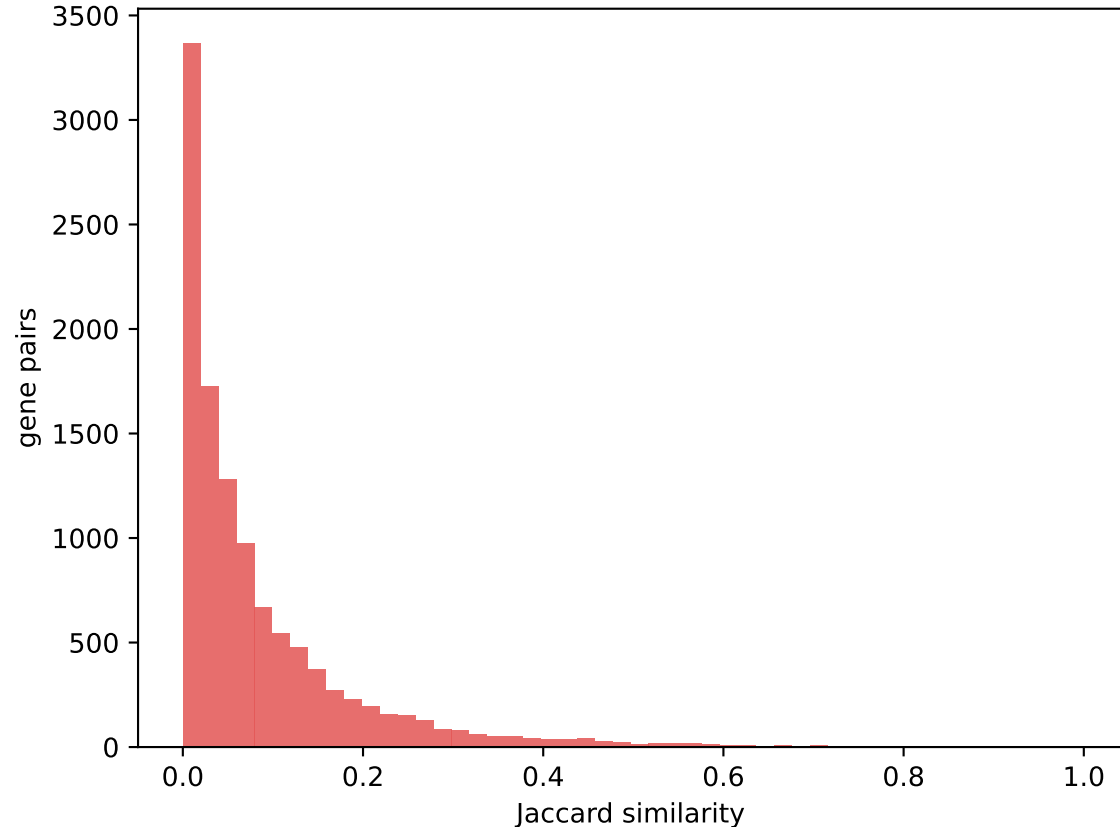
Gene embedding colored by annotation burden



Pairwise gene cosine similarity



Pairwise gene Jaccard similarity



GO terms retained under support/prevalence filters

